

EXPLAINABLE & TRUSTWORTHY AI · PROJECT · POLITECNICO DI TORINO

Characterization of MedConcept Limits to a 2D BiomedCLIP Adaptation.

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THE PROBLEM

Medical VLMs are accurate but opaque.

The internal representations are high-dimensional, polysemic and unreadable. A serious concern in safety-critical clinical use.

§1 Introduction

APPROACH

Concept-based XAI: from supervised to unsupervised discovery.

Supervised

Concept Bottleneck Models (Koh 2020) and TCAV (Kim 2018) need a pre-defined, labelled concept set — exactly the limitation unsupervised discovery targets.

MedConcept (Haque 2026)

SAE extraction → alignment to a radiology vocabulary → an LLM-judge with Aligned/Unaligned/Uncertain metrics.

Impl: `src/autoencoder/`, `src/vocabulary_building/`, `src/evaluate_llm_judge.py`

Sparse Autoencoders

Bricken et al. (2023): dictionary-learning on activations yields monosemantic latents. The Top-K SAE (Gao 2024) sets L_0 exactly via top-k gating. We adopt it.

RESEARCH GAPS · IN THE PAPER'S ORDER

Six gaps this work addresses.

GAP 1

Instability / non-identifiability

Sparse factorisations are not reproducible across seeds; no prior medical-VLM work reports seed-stability against a null.

GAP 3

Representation-location mismatch

SAE literature uses raw hidden states; reference pipelines fit on the already-projected, gap-bearing CLIP space.

GAP 5

Flat concepts

Explanations are top-k lists that ignore anatomical / hierarchical structure.

§3 Research gaps

GAP 2

Clinical validity

A cosine-assigned name is cosmetic unless it tracks real pathology.

GAP 4

Small-data robustness

SAEs need 10^5 – 10^6 activations; licence-free medical corpora are 2–3 orders smaller.

GAP 6

Non-reproducible evaluation

Claims rest on single metrics whose chance floor is never reported; slot-wise overlap is degenerate under per-seed permutation.

BACKBONE & VOCABULARY

Our Adapted Architecture.

Instead of 3D Merlin, we use the 2D BiomedCLIP as backbone. Its image encoder exposes a **768-d hidden state** that a frozen projection compresses into the **512-d** contrastive space. Features are named against **domain-matched** vocabularies: **RadLex** for IU X-Ray (chest-specific), **MeSH** for ROCOV2 (multimodal). Before of this, we correct the **modality gap**.

Impl: `src/embedding_extraction/extract_embeddings.py`, `src/vocabulary_building/build_vocabulary.py`

DATA

Two scales: IU X-Ray and ROCov2 (10×).

IU X-RAY · 1× · PRIMARY

The small-scale setting.

Split **by radiographic study**, so no study appears in both train and test — verified **zero** overlap, recomputed deterministically every run.

ROCOV2 · 10× · REPLICATION

A ten-times
larger corpus.

Split by image. It exists to test one hypothesis head-on: **does more data cure non-identifiability?**

Impl: `src/utils.py` (split_embeddings) · anti-leakage test: `tests/unit/test_split.py`

METHOD · THREE PATHS

Three decompositions, one SAE config.

01 · BASELINE

Top-K SAE · 512-d

Faithful MedConcept re-impl on the projected space. Multi-seed. (Gap 1)

02 · PATH A

Top-K SAE · 768-d

CLS hidden state *before* projection; bridge with frozen W_proj. (Gap 3)

03 · PATH B

SPLiCE

Fixed domain-matched dictionary (RadLex / MeSH) — no learning, deterministic. (Gap 1+4)

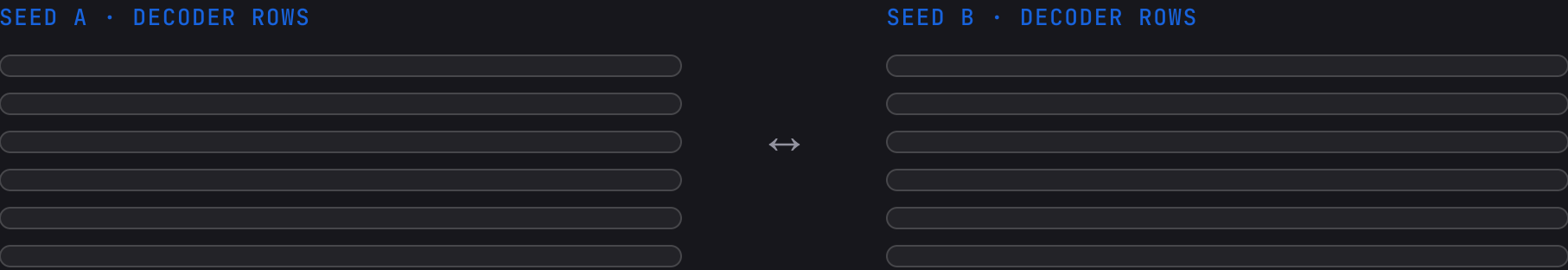
ONE SAE CONFIG FOR CLEAN COMPARISON

DICTIONARY D	TOP-K	STEPS	SEEDS
2048	32	8,000	5 (0, 42, 123, 456, 789)

Impl: `src/autoencoder/sae_module.py` (SAEManager), `src/sae_hidden/`, `src/concept_discovery/spliece.py`

A permutation-invariant matched metric.

For each decoder row of seed A, take its **best cosine match** across *all* rows of seed B, then average. Feature order does not matter — that is what makes it a real stability measure.



Best-match cosine, averaged over rows. Permutation-invariant by construction.

Impl: `src/autoencoder/stability_analysis.py`

A null conditioned on effective rank.

Real decoder directions live in a **subspace** of effective rank ~357 — not the ambient 512. We draw null vectors *inside* that subspace, so the comparison is honest. An isotropic full-512 null would inflate the ratio.

SUBSPACE-CONDITIONED NULL

	IU X-RAY	ROCOV2
Ambient dimensionality	512	512
Decoder effective rank	~357	~363
Null drawn in	top-rank subspace	top-rank subspace

Impl: `src/sae_hidden/naming_hidden.py` (`compute_random_baseline`)

THE HEADLINE RESULT · GAP 1

1.81×

1.67× IUX-Ray · 1.81× ROCov2

The baseline SAE reconstructs its inputs well — but cross-seed **feature identity is absent**: best-match barely above the conditioned null. Genuine non-identifiability, not a training bug.

BASELINE · TAB.1 · GAP 1

Good reconstruction, absent identity.

CROSS-SEED STABILITY, BASELINE TOP-K SAE

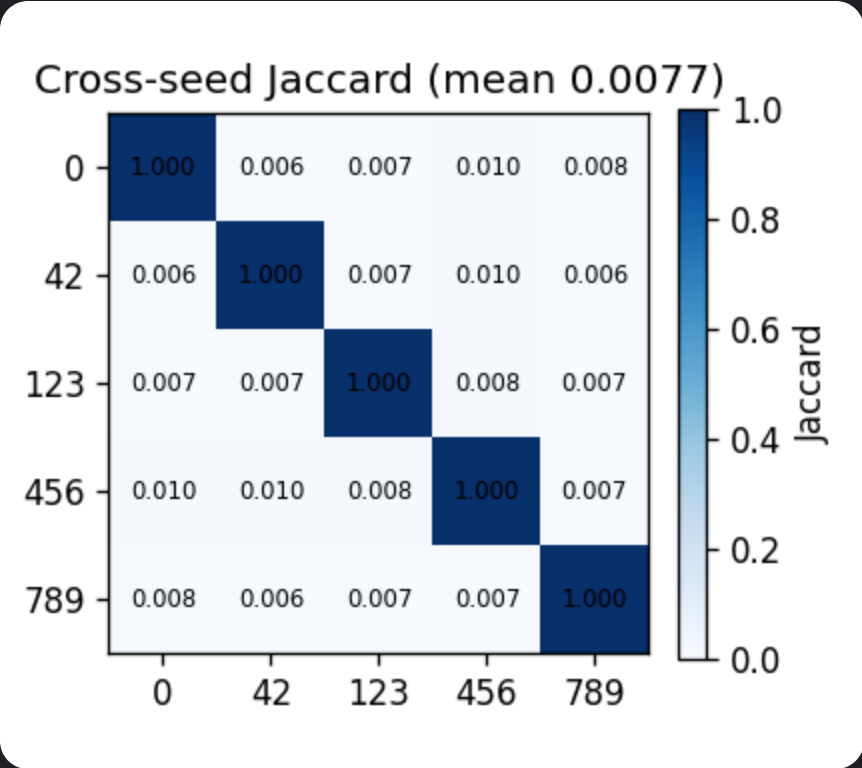
INDICATOR	IU X-RAY	ROCOV2 (10×)
Reconstruction cosine	0.99	0.97
Matched obs / null	1.67×	1.81×
Matched cosine (raw)	0.299	0.327
Features matching ≥0.9	0.0%	0.3%
Naming (top-1 cos)	0.40	0.48

Ref: main_extended.tex §5.1 · Tab.1

Slot-wise is floor-by-construction.

Rename one SAE's features with a random permutation — **same network**, same reconstruction — and slot-wise Jaccard collapses to the **0.0077** floor. The matched metric correctly sees identity.

Impl: `scripts/relabeling_control.py`

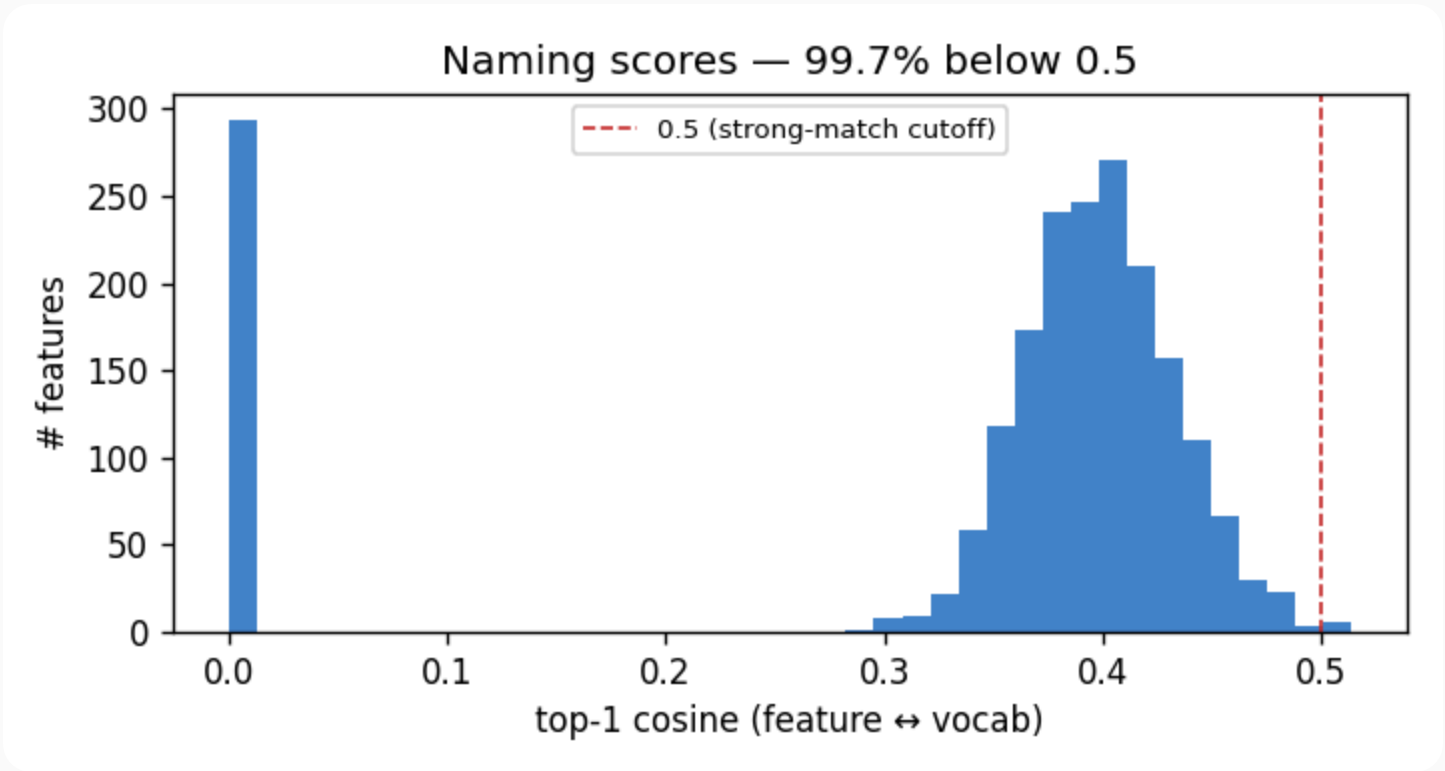


Cross-seed Jaccard (IU baseline). Mean ≈ 0.0077 — the analytical floor.

A healthier decomposition — same verdict.

Path A trains on the hidden state **before** the projection. Fewer dead features, better naming — but the matched verdict is unchanged: **weak universality**, a shared subspace, not identical features. Still the pooled CLS token.

Impl: `src/sae_hidden/`, `scripts/run_path_a.py`



Naming scores — top-1 feature↔vocab cosine. ~99.7% below 0.5 (max 0.513).

PATH A · ABLATION · TAB.3

No overcompleteness escapes weak universality.

MATCHED OBS / NULL VS DICTIONARY OVERCOMPLETENESS

DICTIONARY D	TOP-K	OBS / NULL
1024	16	2.78×
2048	32	2.63×
4096	64	2.00×

More overcomplete → ratio drops toward the null. No setting is identifiable.

Ref: main_extended.tex §5.3 · Tab.3

THE SCALE HYPOTHESIS · GAP 4

10_x

ROCOv2 ~80k vs IU X-Ray ~7k

Ten times more data improves **training health** (dead 16% → 0.6%) but **not identifiability** — matched 0.299 → 0.327, ratio 1.67× → 1.81×. Binding constraint = **representation site**, not volume.

PATH B · SPLICE · GAP 1+4

Deterministic. Judge-ready.

Decompose each embedding into a sparse, non-negative combination over a **fixed domain-matched dictionary** (RadLex / MeSH). Never estimated → **deterministic by construction**, zero cross-seed instability.

- Fixed dictionary — no estimation, no seeds, no instability.
- Same output schema as the SAE — the same judge scores it.
- Honest caveat — frequent terms are "mixed" (a dictionary property).

Impl: `src/concept_discovery/spliece.py` – `decompose_image()`

SPLICE · SCALE + GAP CORRECTION

Scales to both corpora. Gap is the fix.

SPLICE COVERAGE AND MODALITY-GAP CORRECTION

	IU X-RAY	ROCOV2
Dictionary	RadLex · 1031	MeSH · 1024
Terms used	997	1024
Images decomposed	1,515	15,958
Runtime	9.5 s	101 s
Max atom  img cos (gap-corrected)	0.49 → 0.85	0.54 → 0.86

Ref: main_extended.tex §5.4

Dense, but weakly separated.

Cluster the active vocabulary into **concept families** via ontology ancestors (RadLex / MeSH) + two degeneracy guards: leaf-root rejection and a subtree canopy that rejects overly generic ancestors. SPLiCE cuts concepts-per-image ~**1.75x**.

ORGANISATION RESULTS · TAB.4

METHOD	CONCEPTS	FAMILIES	SILHOUETTE
SPLiCE (IU)	997	32	0.020
SAE baseline (IU)	80	9	0.094
Path A hidden (IU)	14	4	0.284
SPLiCE (ROCOv2)	1024	32	0.066

Impl: src/concept_discovery/organize.py · Path A hidden: 1260/1515 images empty

A small upper tail is genuinely faithful.

Do **any** features track real pathology? Correlating activations against the ground-truth labels (with a shuffle null), a **small upper tail** is genuinely faithful — the strongest looks like a *mass*, clinically plausible. But those same features stay **unstable across seeds**: partially faithful, entirely non-reproducible.

Impl: `scripts/run_faithfulness.py` · `main_extended.tex` §5.6

LLM-AS-JUDGE · LOCAL PLAUSIBILITY

81.6%

SPLiCE MedGemma · 79.6% null-k

SPLiCE scores **high local plausibility** under a medical-tuned judge — but **almost nothing** under a general one. **Random-k null explanations score comparably** (79.6%), indicating weak discriminability. The Aligned rate is a property of the **judging model** as much as of the explanation. And this local plausibility is **orthogonal** to global identifiability.

CONCLUSIONS

Three lessons that generalise.

Null + permutation-invariant

Interpretability claims about discovered concepts must be calibrated against analytical nulls **and** a permutation-invariant statistic. Slot-wise overlap is degenerate.

Partially-negative evaluation

An honest, partially-negative evaluation is more useful than a curated one. The defect is localised (representation site), not global.

Ref: `main_extended.tex §6` – 4 contributions: `relabeling control` · `matched metric + null` · `scale refutation` · `partially-negative eval`

Scale is the last suspect

For a factorisation with limited identifiability, data scale is the least likely culprit: 10× changed training health, not identifiability.

LIMITATIONS & FUTURE WORK

SAE on the patch-token residual stream.

The diagnosis is predictive: decomposing BiomedCLIP **patch-token** residual stream (a mid layer, not the pooled CLS) should restore identifiability — the experiment our characterisation points to as the next step.

- Limitation (i): LLM-judge is model-dependent; random-k nulls score comparably.
- Limitation (ii): predicted mitigation is SAE on patch-token residual stream.
- Limitation (iii): ROCov2 confounds scale with domain; PadChest (~160k chest X-rays) is the clean test.

Thank you · questions — Marc'Antonio Lopez · Nicolò Colle · Carmine Francesco Benvenuto